ConstructHub - Business Concept & Software Architecture

1. Executive Summary

ConstructHub is a cloud-native construction management platform connecting organizations, contractors, subcontractors and vendors. It digitizes procurement, inventory, project execution, financial tracking, approvals and document management using an event-driven microservice architecture.

2. Problem Statement

Construction companies often rely on disconnected tools such as spreadsheets, emails, messaging apps and paper-based processes, leading to delays, poor visibility and duplicated work.

3. Business Objectives

Digitize construction workflows, improve collaboration, reduce procurement time, increase transparency, reduce material waste and support multiple organizations and projects.

4. Stakeholders

Organization Administrator, Project Manager, Constructor, Vendor.

5. Functional Modules

User Management, User Preferences, Membership & Roles, Authorization, Organizations, Projects, SubProjects, Vendor Management, Materials & Inventory, Procurement, Delivery Management, Financials, Workflow & Approvals, Notifications, Subcontractor Progress, Reviews, Document Management, Reporting & Analytics.

6. Business Processes

Procurement: Request → Approval → Vendor → Delivery → Inventory → Financials. Material Delivery: Vendor → Delivery → Site Confirmation → Inventory → Financials. User Assignment: Organization → Project → SubProject → User → Role → Permissions.

7. Domain Model

Organization → Project → SubProject → Membership → User → Role → Permissions.

8. Software Architecture

Domain-Driven Design (DDD), CQRS, Event Sourcing, Event Stream, Read Models, OAuth2 and Microservices.

9. Event Flow

ProcurementCreated → Workflow → Vendor → Notification → Delivery → Inventory → Financials.

10. Security

OAuth2, JWT, RBAC, organization isolation, project isolation and audit logs.

11. Technology Stack

Frontend: React & TypeScript. Backend: Java & Spring Boot. Messaging: Kafka. Database: PostgreSQL. Storage: S3-compatible. Authentication: Keycloak. Deployment: Kubernetes.

12. Roadmap

Phase 1: Users, Organizations, Projects. Phase 2: Vendors, Procurement, Inventory. Phase 3: Deliveries, Financials, Workflow. Phase 4: Reporting & AI Document Analysis.

13. Business Model

Subscription plans: Starter, Professional and Enterprise. Customers include construction companies, general contractors and vendors.

14. Future Features

Mobile apps, BIM integration, IoT sensors, equipment tracking, AI risk prediction and AI cost forecasting.

Recommendation

Organize the platform around business capabilities: Identity & Access, Project Management, Supply Chain, Financial Management, Construction Execution, Collaboration and Reporting.